



NORTH SYDNEY GIRLS HIGH SCHOOL
YEAR 12 – TERM 2 ASSESSMENT
2007

MATHEMATICS
EXTENSION COURSE 2

TIME ALLOWED: 60 minutes
Plus 2 minutes reading time

INSTRUCTIONS:

- Start each question on a new page
- Hand each question in separately, including a sheet for non-attempts
- Show all necessary working

This task is worth 35% of the HSC Assessment Mark

Question 1 (20 Marks)

- (a) Use implicit differentiation to find the value of $\frac{dy}{dx}$ at the point (1,-2) on

$$(x+y)^3 = -x$$

3

(b)

- i. Express $x^2 - 4x + 6$ in the form $(x-a)^2 + b$

1

- ii. Consider the implicit function

$$2x^2y - 8xy + 12y + 1 = 0$$

Express 'y' in terms of 'x' ie $y = f(x)$

2

- iii. By writing the denominator of $f(x)$ in part (ii) in the form in part (i), ie $(x-a)^2 + b$, find the domain and range of $y = f(x)$.

2

(c) Given the equation

$$2x^2 + 2y^2 = xy + 30$$

- i. Show that $\frac{dy}{dx}(4y-x) = y-4x$

2

- ii. Find the respective points where the curve has horizontal tangents AND where it has vertical tangents

4

- iii. Use the information from part (ii) to help you sketch the graph of the function, showing its important features.

2

(d) The equation $|z+4-2i| - |z-2-2i| = \pm 1$ corresponds to a hyperbola.

- i. Write down the complex number which represents the centre of the hyperbola.
ii. State the lengths of its semi transverse and semi conjugate axes.

1**3****Question 2 (20 Marks)**

- (a) The three roots of $x^3 + px + q = 0$ are given by α, β and γ .

3

Find a cubic equation whose roots are α^2, β^2 and γ^2 .

- (b) (i) If $(x-r)^2$ is a factor of the polynomial $P(x)$,
prove that $(x-r)$ is a factor of $P'(x)$.

3

- (ii) The polynomial $P(x) = x^4 + bx^3 + cx^2 - 24x + 36$ has a double zero at $x = 2$.
Determine the values of b and c and hence write $P(x)$ as a product of four linear factors.

6

- (c) (i) Use De Moivre's theorem to find the roots of $z^5 + 1 = 0$ showing that they are equally spaced on a unit circle in an Argand Diagram. 2
- (ii) Express $z^5 + 1$ as a product of irreducible factors with real coefficients. 3
- (iii) Show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ and $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$ 3

Question 3 (20 Marks)

- a) i Find the real values of k for which the equation

$$\frac{x^2}{6-k} + \frac{y^2}{4-k} = 1$$

defines respectively an ellipse and a hyperbola. 3

- ii Sketch the curve for $k = 5$ showing its important features. 5
- iii Explain how the shape of this graph changes as k increases from 5 to 6 and determine if the graph has a limiting position for this increase. 2

- b) $P(a \cos \theta, b \sin \theta)$ where $0 < \theta < \frac{\pi}{2}$, is a point which lies on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

An interval is drawn from O the origin, parallel to the tangent at P , to meet the ellipse at Q .

- i Show that the equation of OQ is given by $y = -\frac{b \cos \theta}{a \sin \theta} x$ 3
- ii Show that there are two possible positions of Q given by $(a \sin \theta, -b \cos \theta)$ and $(-a \sin \theta, b \cos \theta)$. 3
- iii Let M be the foot of the perpendicular from P to OQ .
Hence, or otherwise, prove that the area of $\triangle OPQ$ is independent of θ . 4

End of paper

$$1a) (x+y)^3 = -x$$

$$3(x+y)^2(1+y') = -1 \quad (1)$$

$$1+y' = -\frac{1}{3(x+y)^2}$$

$$y' = -\frac{1}{3(x+y)^2} - 1 \quad (1)$$

At $x=1, y=-2$

$$y' = -\frac{1}{3(-1)^2} - 1$$

$$= -\frac{1}{3} - 1$$

$$= -\frac{4}{3}$$

$$b) (i) x^2 - 4x + 6 = (x^2 - 4x + 4) + 2$$

$$= (x-2)^2 + 2 \quad (1)$$

$$(ii) 2x^2y - 8xy + 12y + 1 = 0$$

$$y(2x^2 - 8x + 12) = -1$$

$$y = -\frac{1}{2(x^2 - 4x + 6)}$$

$$y = -\frac{1}{2(x-2)^2 + 2}$$

$$(iii) D = \{x: x \in \mathbb{R}, \}$$

$$R = \{y: -\frac{1}{4} \leq y \leq 0\}$$

$$c) 2x^2 + 2y^2 = xy + 30$$

$$4x + 4yy' = x + y$$

$$4yy' - xy' = y - 4x$$

$$y'(4y - x) = y - 4x$$

$$y' = \frac{y-4x}{4y-x}$$

Horizontal tangents
at $y-4x=0$

$$y=4x$$

$$2x^2 + 2(4x)^2 = xy + 30$$

$$36x^2 = 4x^2 + 30$$

$$30x^2 = 30$$

SOLNS EXT 2 ASSESSMENT
TASK, T2 2007, N.S.G.H.S

$$x^2 = 1 \Rightarrow x = \pm 1$$

$(1, 4)(-1, -4)$ have horizontal tangents

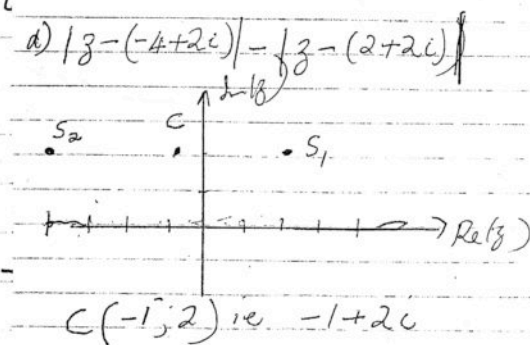
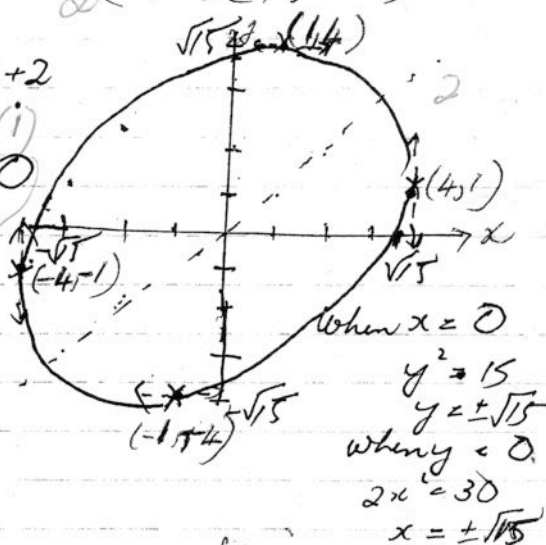
Vertical Tangents
at $4y=x$

$$2(4y) + 2y^2 = 4y + 30$$

$$32y^2 + 2y^2 = 4y + 30$$

$$y = \pm 1$$

$(4, 1)(-4, -1)$



$$d) |3 - (-4+2i)| = |3 - (2+2i)|$$

$$2a = 1 \quad b^2 = a^2(e^2 - 1)$$

$$a = \frac{1}{2} \quad b^2 = \frac{1}{4}(36 - 1)$$

$$b^2 = \frac{35}{4}$$

$$ae = 3$$

$$\frac{1}{2}e = 3$$

$$e = 6$$

LENGTH OF SEMI TRANSVERSE AXIS

LENGTH OF SEMI CONJUGATE AXIS

Question 2

$$(a) \quad x^3 + px + q = 0 \Rightarrow x(x^2 + p) = -q$$

$$\therefore x^2(x^2 + p)^2 = q^2$$

$$\text{Let } y = x^2$$

$$\therefore y(y + p)^2 = q^2$$

$$\therefore y^3 + 2py^2 + p^2y - q^2 = 0$$

Alternatively

$$\text{Let } y = x^2 \Rightarrow x = \sqrt{y}$$

$$\therefore x^3 + px + q = 0 \Rightarrow (\sqrt{y})^3 + p\sqrt{y} + q = 0$$

$$\therefore y\sqrt{y} + p\sqrt{y} = -q$$

$$\therefore \sqrt{y}(y + p) = -q \Rightarrow [\sqrt{y}(y + p)]^2 = q^2$$

$$\therefore y(y^2 + 2py + p^2) = q^2$$

$$\therefore y^3 + 2py^2 + p^2y - q^2 = 0$$

$$(b) \quad (i) \quad P(x) = (x - r)^2 Q(x), \quad Q(r) \neq 0$$

$$\therefore P'(x) = 2(x - r)Q(x) + (x - r)^2 Q'(x)$$

$$\therefore P(x) = (x - r)[2Q(x) + (x - r)Q'(x)]$$

$$(ii) \quad P(x) = x^4 + bx^3 + cx^2 - 24x + 36 = (x - 2)^2 Q(x)$$

$$\therefore P(2) = P'(2) = 0$$

$$P'(x) = 4x^3 + 3bx^2 + 2cx - 24$$

$$P(2) = 16 + 8b + 4c - 48 + 36 = 0$$

$$\therefore 8b + 4c + 4 = 0$$

$$\therefore 2b + c = -1 \quad -(1)$$

$$P'(2) = 32 + 12b + 4c - 24 = 0$$

$$\therefore 12b + 4c + 8 = 0$$

$$\therefore 3b + c = -2 \quad -(2)$$

$$(2) - (1) \Rightarrow b = -1$$

$$\therefore c = 1$$

$$\therefore b = -1, c = 1$$

$$\therefore P(x) = x^4 - x^3 + x^2 - 24x + 36 = (x - 2)^2 (x^2 + Dx + 9)$$

$$\text{Substitute } x = 1: \quad 1 - 1 + 1 - 24 + 36 = 1(10 + D)$$

$$\therefore D = 3$$

$$\therefore P(x) = x^4 - x^3 + x^2 - 24x + 36 = (x - 2)^2 (x^2 + 3x + 9)$$

$$\text{Now } x^2 + 3x + 9 = x^2 + 3x + 2\frac{1}{4} + 8\frac{3}{4} = \left(x + \frac{3}{2}\right)^2 + \frac{27}{4} = \left(x + \frac{3}{2}\right)^2 - \frac{27}{4}i^2$$

$$\therefore x^2 + 3x + 9 = \left(x + \frac{3}{2} + i\frac{3\sqrt{3}}{2}\right)\left(x + \frac{3}{2} - i\frac{3\sqrt{3}}{2}\right)$$

$$\begin{aligned} P(x) &= (x-2)(x-2)\left(x + \frac{3}{2} + i\frac{3\sqrt{3}}{2}\right)\left(x + \frac{3}{2} - i\frac{3\sqrt{3}}{2}\right) \\ &= \frac{1}{4}(x-2)(x-2)(2x+3+i3\sqrt{3})(2x+3-i3\sqrt{3}) \end{aligned}$$

(c) (i) $z^5 = -1 = \text{cis}(\pi + 2k\pi) = \text{cis}(2k+1)\pi, k \in \mathbb{Z}$

$$\therefore z = \text{cis}\left(\frac{2k+1}{5}\pi\right) \quad [\text{de Moivre's Theorem}]$$

$$z = \text{cis}\left(\frac{\pi}{5} + \frac{2\pi k}{5}\right)$$

$\therefore |z| = 1$ and $\arg z$ are $\frac{2\pi}{5}$ radians apart. So the solutions are equally spaced around a unit circle.

$$\therefore z = \text{cis}\left(\frac{\pi}{5}\right), \text{cis}\left(\frac{3\pi}{5}\right), \text{cis}\left(\frac{5\pi}{5}\right), \text{cis}\left(\frac{7\pi}{5}\right), \text{cis}\left(\frac{9\pi}{5}\right)$$

$$\therefore z = -1, \text{cis}\left(\frac{\pi}{5}\right), \text{cis}\left(-\frac{\pi}{5}\right), \text{cis}\left(\frac{3\pi}{5}\right), \text{cis}\left(-\frac{3\pi}{5}\right)$$

(ii)
$$\begin{aligned} z^5 + 1 &= (z+1)\left[z - \text{cis}\left(\frac{\pi}{5}\right)\right]\left[z - \text{cis}\left(-\frac{\pi}{5}\right)\right]\left[z - \text{cis}\left(\frac{3\pi}{5}\right)\right]\left[z - \text{cis}\left(-\frac{3\pi}{5}\right)\right] \\ &= (z+1)\left(z^2 - 2z\cos\frac{\pi}{5} + 1\right)\left(z^2 - 2z\cos\frac{3\pi}{5} + 1\right) \\ &\quad \left[\because (z-\alpha)(z-\bar{\alpha}) = z^2 - 2z\text{Re}\alpha + |\alpha|^2\right] \end{aligned}$$

(iii) Now $-1 + \text{cis}\left(\frac{\pi}{5}\right) + \text{cis}\left(-\frac{\pi}{5}\right) + \text{cis}\left(\frac{3\pi}{5}\right) + \text{cis}\left(-\frac{3\pi}{5}\right) = 0$ (sum of roots of $z^5 + 1 = 0$)

$$\therefore 2\cos\frac{\pi}{5} + 2\cos\frac{3\pi}{5} = -1 \quad (\alpha + \bar{\alpha} = 2\text{Re}\alpha)$$

$$\therefore \cos\frac{\pi}{5} + \cos\frac{3\pi}{5} = -\frac{1}{2}$$

Substitute $z = i$ into $z^5 + 1 = (z + 1)\left(z^2 - 2z \cos \frac{\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right)$

$$\therefore i^5 + 1 = (i + 1)\left(-2i \cos \frac{\pi}{5}\right)\left(-2i \cos \frac{3\pi}{5}\right)$$

$$\therefore 1 = -4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$$

$$\therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$$

Alternative 1

$$\text{Using } z^5 + 1 = (z + 1)\left(z^2 - 2z \cos \frac{\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right)$$

Equating the coefficients of z of both sides obtains:

$$0 = 1 - 2 \cos \frac{\pi}{5} - 2 \cos \frac{3\pi}{5}$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2} \quad (1)$$

Equating the coefficients of z^2 of both sides obtains:

$$0 = 1 + 1 + 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} - 2 \cos \frac{\pi}{5} - 2 \cos \frac{3\pi}{5}$$

$$\therefore 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = 2 \left(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} \right) - 2 = -1 \quad [\text{from (1)}]$$

$$\therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$$

Alternative 2

$$\begin{aligned} z^5 + 1 &= (z + 1)(1 - z + z^2 - z^3 + z^4) \\ &= (z + 1)\left(z^2 - 2z \cos \frac{\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right) \end{aligned}$$

$$\therefore 1 - z + z^2 - z^3 + z^4 = \left(z^2 - 2z \cos \frac{\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right)$$

Equating the coefficients of z of both sides obtains:

$$-2 \cos \frac{\pi}{5} - 2 \cos \frac{3\pi}{5} = -1$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

Equating the coefficients of z^2 of both sides obtains:

$$1 + 1 + 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = 1$$

$$\therefore 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -1$$

$$\therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$$

① 3

a) Ellipse when both

$$6-k > 0 \Rightarrow k < 6$$

and $4-k > 0 \Rightarrow k < 4$

ie $k < 4 \Rightarrow$ Ellipse

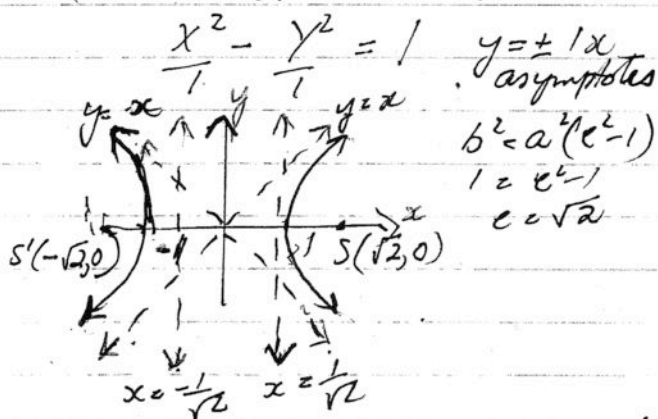
Hyperbola when both

$$6-k > 0 \Rightarrow k < 6$$

$$\text{AND } 4-k < 0 \Rightarrow k > 4$$

ie $4 < k < 6$

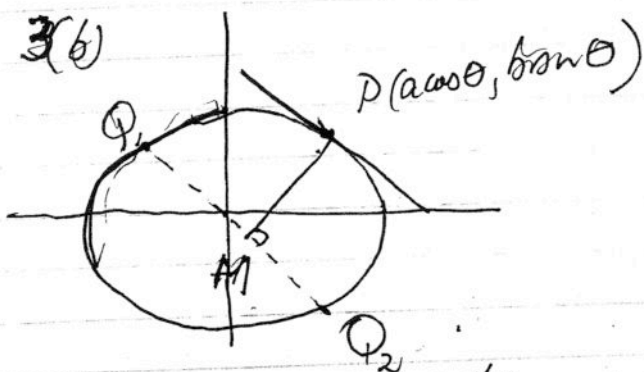
(i) $k=5$



(iii) As b' is approached by k , a^2 becomes small.

eg $\therefore \frac{x^2}{\frac{1}{1000}} + \frac{y^2}{-2} = 1$ ie $\frac{x^2}{(\frac{1}{1000})} - \frac{y^2}{2} = 1$ This shows the length of the transverse axis will approach zero

a) The length of the transverse axis approaches zero. Hence the branches become flatter, approaching a limiting position which is the y-axis



$$(1) \quad x = a \cos \theta \Rightarrow \frac{dx}{d\theta} = -a \sin \theta$$

$$y = b \sin \theta \Rightarrow \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{b \cos \theta}{-a \sin \theta}$$

Since OP is $\perp$ to the tangent $y = mx$

$$y = -\frac{b \cos \theta}{a \sin \theta} x$$

It solve

$$(11) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (1)$$

$$y = -\frac{b \cos \theta}{a \sin \theta} x \quad (2)$$

simultaneously

$$\frac{x^2}{a^2} + \frac{\left(-\frac{b \cos \theta}{a \sin \theta} x\right)^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{x^2 \cos^2 \theta}{a^2 \sin^2 \theta} = 1$$

$$(x) \quad a^2 \sin^2 \theta$$

$$a^2 \sin^2 \theta + a^2 \cos^2 \theta = a^2 \sin^2 \theta$$

$$x^2 = a^2 \sin^2 \theta$$

$$x = \pm a \sin \theta$$

By substitution into (1)

$$(2) \text{ becomes } x b \cos \theta + y a \sin \theta = 0$$

$$Q_1(a \sin \theta, -b \cos \theta)$$

$$Q_2(-a \sin \theta, b \cos \theta)$$

$$(111) \quad \text{Area} = \frac{1}{2} PM \cdot OQ_2$$

$$PM = \frac{|b \cos \theta \cdot a \cos \theta + a \sin \theta \cdot b \sin \theta|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \frac{ab(\cos^2 \theta + \sin^2 \theta)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$OQ_2 = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$A = \frac{1}{2} \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$* \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$A = \frac{1}{2} ab$ and a, b are independent of θ